

```
In [265]: txt = "  abc def ghi  "
          txt.lstrip()
```

```
Out[265]: 'abc def ghi  '
```

```
In [266]: txt = "  abc def ghi  "
          txt.strip()
```

```
Out[266]: 'abc def ghi'
```

Using Escape Character

```
In [252]: #Using double quotes in the string is not allowed.
          mystr = "My favourite TV Series is "Game of Thrones""

          File "<ipython-input-252-0fa35a74da86>", line 2
            mystr = "My favourite TV Series is "Game of Thrones""
                                     ^
          SyntaxError: invalid syntax
```

```
In [253]: #Using escape character to allow illegal characters
          mystr = "My favourite series is \"Game of Thrones\""
          print(mystr)

          My favourite series is "Game of Thrones"
```

List

- 1) List is an ordered sequence of items.
- 2) We can have different data types under a list. E.g we can have integer, float and string items in a same list.

List Creation

```
In [423]: list1 = []           # Empty List
```

```
In [491]: print(type(list1))

          <class 'list'>
```

```
In [424]: list2 = [10,30,60]    # List of integers numbers
```

```
In [425]: list3 = [10.77,30.66,60.89]    # List of float numbers
```

```
In [426]: list4 = ['one', 'two' , "three"]    # List of strings
```

```
In [427]: list5 = ['Asif', 25 , [50, 100], [150, 90]]    # Nested Lists
```

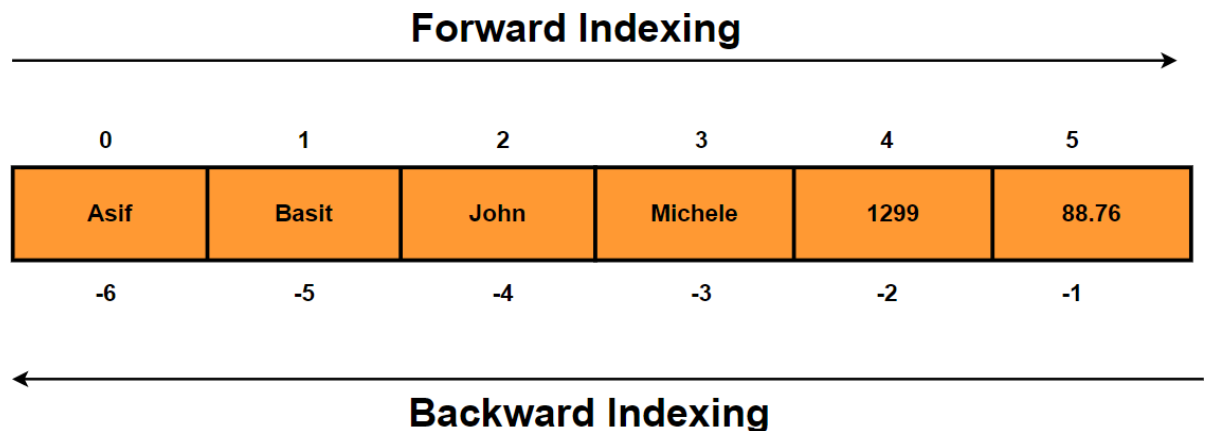
```
In [428]: list6 = [100, 'Asif', 17.765]    # List of mixed data types
```

```
In [429]: list7 = ['Asif', 25 , [50, 100], [150, 90] , {'John' , 'David'}]
```

```
In [430]: len(list6) #Length of list
```

```
Out[430]: 3
```

List Indexing



```
In [432]: list2[0] # Retrieve first element of the list
```

```
Out[432]: 10
```

```
In [433]: list4[0] # Retrieve first element of the list
```

```
Out[433]: 'one'
```

```
In [434]: list4[0][0] # Nested indexing - Access the first character of the first list element
```

```
Out[434]: 'o'
```

```
In [435]: list4[-1] # Last item of the list
```

```
Out[435]: 'three'
```

```
In [436]: list5[-1] # Last item of the list
```

```
Out[436]: [150, 90]
```

List Slicing

```
In [437]: mylist = ['one' , 'two' , 'three' , 'four' , 'five' , 'six' , 'seven' , 'eight']
```

```
In [438]: mylist[0:3] # Return all items from 0th to 3rd index location excluding the item
```

```
Out[438]: ['one', 'two', 'three']
```

```
In [439]: mylist[2:5] # List all items from 2nd to 5th index location excluding the item a
```

```
Out[439]: ['three', 'four', 'five']
```

```
In [440]: mylist[:3] # Return first three items
```

```
Out[440]: ['one', 'two', 'three']
```

```
In [441]: mylist[:2] # Return first two items
```

```
Out[441]: ['one', 'two']
```

```
In [442]: mylist[-3:] # Return Last three items
```

```
Out[442]: ['six', 'seven', 'eight']
```

```
In [443]: mylist[-2:] # Return Last two items
```

```
Out[443]: ['seven', 'eight']
```

```
In [444]: mylist[-1] # Return Last item of the List
```

```
Out[444]: 'eight'
```

```
In [445]: mylist[:] # Return whole List
```

```
Out[445]: ['one', 'two', 'three', 'four', 'five', 'six', 'seven', 'eight']
```

Add , Remove & Change Items

```
In [446]: mylist
```

```
Out[446]: ['one', 'two', 'three', 'four', 'five', 'six', 'seven', 'eight']
```

```
In [447]: mylist.append('nine') # Add an item to the end of the List  
mylist
```

```
Out[447]: ['one', 'two', 'three', 'four', 'five', 'six', 'seven', 'eight', 'nine']
```

```
In [448]: mylist.insert(9,'ten') # Add item at index location 9  
mylist
```

```
Out[448]: ['one', 'two', 'three', 'four', 'five', 'six', 'seven', 'eight', 'nine', 'ten']
```

```
In [449]: mylist.insert(1,'ONE') # Add item at index Location 1
mylist
```

```
Out[449]: ['one',
           'ONE',
           'two',
           'three',
           'four',
           'five',
           'six',
           'seven',
           'eight',
           'nine',
           'ten']
```

```
In [450]: mylist.remove('ONE') # Remove item "ONE"
mylist
```

```
Out[450]: ['one', 'two', 'three', 'four', 'five', 'six', 'seven', 'eight', 'nine', 'ten']
```

```
In [451]: mylist.pop() # Remove Last item of the List
mylist
```

```
Out[451]: ['one', 'two', 'three', 'four', 'five', 'six', 'seven', 'eight', 'nine']
```

```
In [452]: mylist.pop(8) # Remove item at index Location 8
mylist
```

```
Out[452]: ['one', 'two', 'three', 'four', 'five', 'six', 'seven', 'eight']
```

```
In [453]: del mylist[7] # Remove item at index Location 7
mylist
```

```
Out[453]: ['one', 'two', 'three', 'four', 'five', 'six', 'seven']
```

```
In [454]: # Change value of the string
mylist[0] = 1
mylist[1] = 2
mylist[2] = 3
mylist
```

```
Out[454]: [1, 2, 3, 'four', 'five', 'six', 'seven']
```

```
In [455]: mylist.clear() # Empty List / Delete all items in the List
mylist
```

```
Out[455]: []
```

```
In [456]: del mylist # Delete the whole list
mylist
```

```
-----
NameError                                Traceback (most recent call last)
<ipython-input-456-50c7849aa2cb> in <module>
      1 del mylist # Delete the whole list
----> 2 mylist
```

NameError: name 'mylist' is not defined

Copy List

```
In [457]: mylist = ['one', 'two', 'three', 'four', 'five', 'six', 'seven', 'eight', 'nine']
```

```
In [458]: mylist1 = mylist # Create a new reference "mylist1"
```

```
In [459]: id(mylist) , id(mylist1) # The address of both mylist & mylist1 will be the same
```

```
Out[459]: (1537348392776, 1537348392776)
```

```
In [460]: mylist2 = mylist.copy() # Create a copy of the list
```

```
In [461]: id(mylist2) # The address of mylist2 will be different from mylist because mylis
```

```
Out[461]: 1537345955016
```

```
In [462]: mylist[0] = 1
```

```
In [463]: mylist
```

```
Out[463]: [1, 'two', 'three', 'four', 'five', 'six', 'seven', 'eight', 'nine', 'ten']
```

```
In [464]: mylist1 # mylist1 will be also impacted as it is pointing to the same list
```

```
Out[464]: [1, 'two', 'three', 'four', 'five', 'six', 'seven', 'eight', 'nine', 'ten']
```

```
In [465]: mylist2 # Copy of list won't be impacted due to changes made on the original lis
```

```
Out[465]: ['one', 'two', 'three', 'four', 'five', 'six', 'seven', 'eight', 'nine', 'ten']
```

Join Lists

```
In [466]: list1 = ['one', 'two', 'three', 'four']
list2 = ['five', 'six', 'seven', 'eight']
```

```
In [467]: list3 = list1 + list2 # Join two lists by '+' operator
list3
```

```
Out[467]: ['one', 'two', 'three', 'four', 'five', 'six', 'seven', 'eight']
```

```
In [468]: list1.extend(list2) #Append list2 with list1  
list1
```

```
Out[468]: ['one', 'two', 'three', 'four', 'five', 'six', 'seven', 'eight']
```

List Membership

```
In [469]: list1
```

```
Out[469]: ['one', 'two', 'three', 'four', 'five', 'six', 'seven', 'eight']
```

```
In [470]: 'one' in list1 # Check if 'one' exist in the list
```

```
Out[470]: True
```

```
In [471]: 'ten' in list1 # Check if 'ten' exist in the list
```

```
Out[471]: False
```

```
In [472]: if 'three' in list1: # Check if 'three' exist in the list  
           print('Three is present in the list')  
else:  
    print('Three is not present in the list')
```

Three is present in the list

```
In [473]: if 'eleven' in list1: # Check if 'eleven' exist in the list  
           print('eleven is present in the list')  
else:  
    print('eleven is not present in the list')
```

eleven is not present in the list

Reverse & Sort List

```
In [474]: list1
```

```
Out[474]: ['one', 'two', 'three', 'four', 'five', 'six', 'seven', 'eight']
```

```
In [475]: list1.reverse() # Reverse the list  
list1
```

```
Out[475]: ['eight', 'seven', 'six', 'five', 'four', 'three', 'two', 'one']
```

```
In [476]: list1 = list1[::-1] # Reverse the list  
list1
```

```
Out[476]: ['one', 'two', 'three', 'four', 'five', 'six', 'seven', 'eight']
```

```
In [477]: mylist3 = [9,5,2,99,12,88,34]
mylist3.sort()    # Sort list in ascending order
mylist3
```

```
Out[477]: [2, 5, 9, 12, 34, 88, 99]
```

```
In [478]: mylist3 = [9,5,2,99,12,88,34]
mylist3.sort(reverse=True)    # Sort list in descending order
mylist3
```

```
Out[478]: [99, 88, 34, 12, 9, 5, 2]
```

```
In [584]: mylist4 = [88,65,33,21,11,98]
sorted(mylist4)    # Returns a new sorted list and doesn't change original L
```

```
Out[584]: [11, 21, 33, 65, 88, 98]
```

```
In [585]: mylist4
```

```
Out[585]: [88, 65, 33, 21, 11, 98]
```

Loop through a list

```
In [481]: list1
```

```
Out[481]: ['one', 'two', 'three', 'four', 'five', 'six', 'seven', 'eight']
```

```
In [482]: for i in list1:
          print(i)
```

```
one
two
three
four
five
six
seven
eight
```

```
In [483]: for i in enumerate(list1):
          print(i)
```

```
(0, 'one')
(1, 'two')
(2, 'three')
(3, 'four')
(4, 'five')
(5, 'six')
(6, 'seven')
(7, 'eight')
```

Count


```
In [485]: list10 = ['one', 'two', 'three', 'four', 'one', 'one', 'two', 'three']
```

```
In [486]: list10.count('one') # Number of times item "one" occurred in the list.
```

```
Out[486]: 3
```

```
In [487]: list10.count('two') # Occurrence of item 'two' in the list
```

```
Out[487]: 2
```

```
In [489]: list10.count('four') #Occurrence of item 'four' in the list
```

```
Out[489]: 1
```

All / Any

The **all()** method returns:

- **True** - If all elements in a list are true
- **False** - If any element in a list is false

The **any()** function returns True if any element in the list is True. If not, any() returns False.

```
In [816]: L1 = [1,2,3,4,0]
```

```
In [817]: all(L1) # Will Return false as one value is false (Value 0)
```

```
Out[817]: False
```

```
In [818]: any(L1) # Will Return True as we have items in the list with True value
```

```
Out[818]: True
```

```
In [819]: L2 = [1,2,3,4,True,False]
```

```
In [820]: all(L2) # Returns false as one value is false
```

```
Out[820]: False
```

```
In [821]: any(L2) # Will Return True as we have items in the list with True value
```

```
Out[821]: True
```

```
In [822]: L3 = [1,2,3,True]
```

```
In [823]: all(L3) # Will return True as all items in the list are True
```

```
Out[823]: True
```